

TURN
YOUR WORLD
INTO A SONG

REAL SOUNDS
INFINITE MUSIC



ANYTHING CAN BE A STUDIO



CUPS



PLANTS



LAPTOPS



BOTTLES



KEYS

Scan your surroundings, capture real sounds,
and let AI turn them into music.

Get Started



See How It Works



THE PROBLEM

Music apps shut most people out

01

You need gear and training

Instruments, sample packs and theory come before the first beat. Most people never get there.

02

Built for eyes only

Timelines, piano rolls and tiny icons leave blind and low-vision users with little they can use on their own.

03

All music lives in the cloud

Your room, your voice and your ideas get uploaded to a server, and nothing works offline.

THE SOLUTION

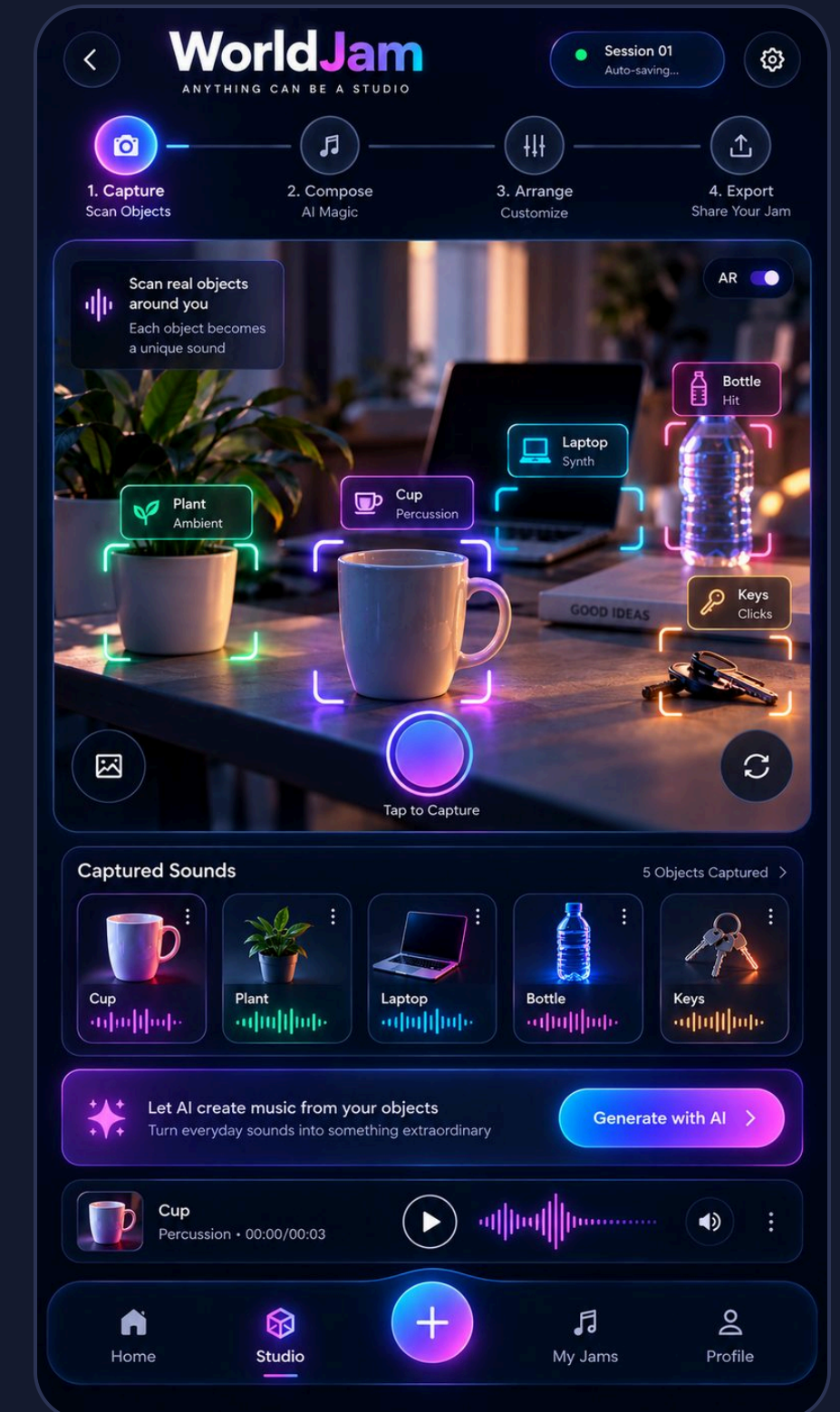
Point at a cup. Record it. Play it.

Tap anything around you and WorldJam keeps its real sound. You program a beat with it, and on-device AI produces a full track around your beat.

Real sounds — your objects are the drums. No sample packs.

On-device AI — Gemma 4 plans the song, offline.

Audio-first — the camera names what it sees, out loud.



Capture screen (design)

HOW IT WORKS

Five steps from room to record

1

See

The camera spots objects and says the main one aloud.

NEW · in
development

2

Capture

Hold, strike it, and its real sound is kept and named.

3

Beat

Tap an 8 or 16-step grid, or record the beat live.

4

Produce

Gemma keeps your beat and builds the rest of the track.

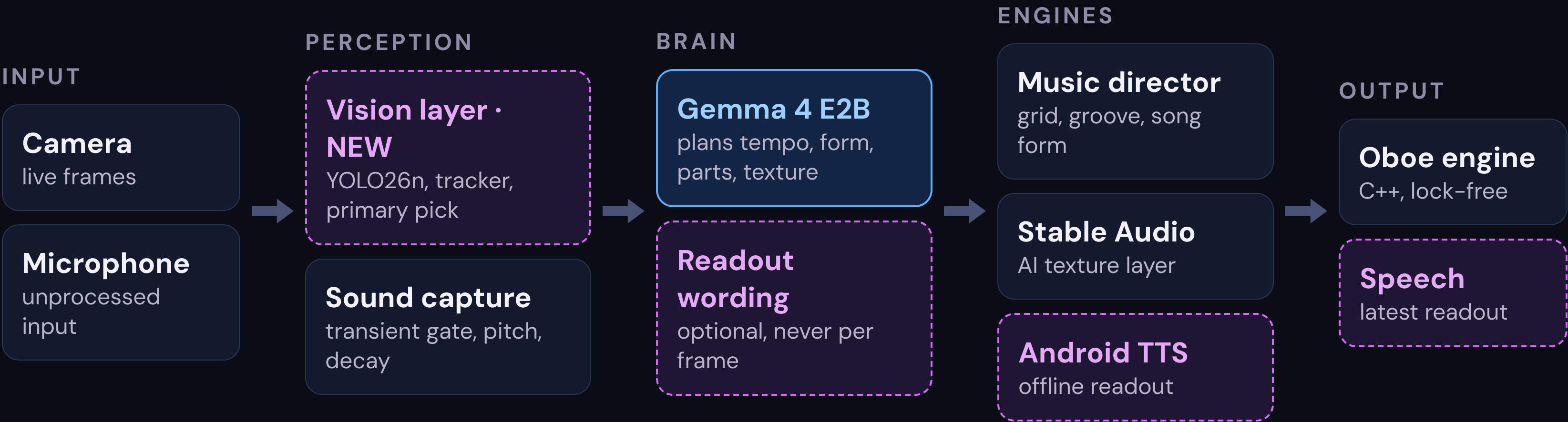
5

Share

Save the jam and export it as a WAV track.

ARCHITECTURE

One on-device pipeline, two senses



Solid = running today · dashed = Phase 1 vision + readout, in development · no cloud vision, LLM or TTS

Three models, one job each

Model	Job	Size	Runs on	Status
Gemma 4 E2B (Q4_O)	Music director, the brain	3.35 GB	llama.cpp (llama.rn), CPU	Working
Stable Audio Open Small (Q5)	AI texture layer	436 MB	sa3.cpp (ggml), own process	Runs on iQOO 15; app build in progress
YOLO26n	Object detection for readout	nano	QNN on Hexagon NPU, LiteRT fallback	In development

Every model has a floor. If one is missing, slow or wrong, a deterministic rule-based arranger keeps the music going, and Qwen3-1.7B is a supported stand-in brain.

THE MUSIC DIRECTOR

You write the beat. The AI produces around it.

YOUR BEAT · 8 STEPS

Cup	■	■	■	■	■	■	■
Table	■	■	■	■	■	■	■
Keys	■	■	■	■	■	■	■
Bottle	□	□	□	□	□	□	□

Empty row: Gemma writes this part

Your hits are fixed

Every cell you light plays, in every section. The model cannot rewrite it.

AI Producer fills the rest

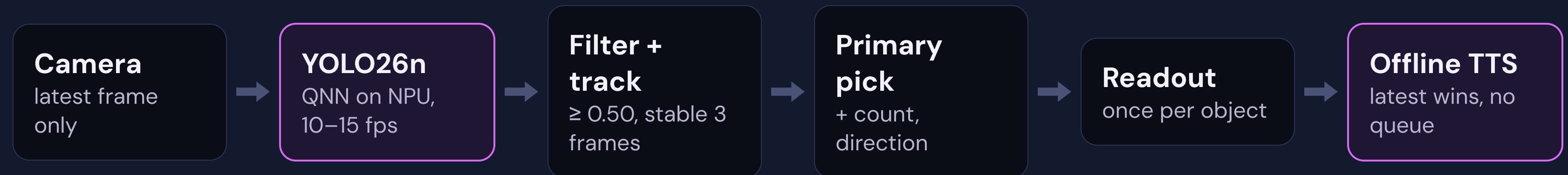
Parts for the empty rows, bass, chords, an 8-bar form and a generated texture.

It plays like a musician

Genre tempo ranges, swing, humanised timing, and ringing objects kept off their own tails.

NEW LAYER · PHASE 1 · IN DEVELOPMENT

Audio-first object awareness



WHAT THE USER HEARS

"Laptop detected."

"Laptop on your right. Two other objects are visible."

RULES

🔊 One big button repeats the latest readout, with no new inference.

Readout toggle: off stops automatic speech, repeat still works.

Works with Gemma off. Never speaks scores or IDs.

MANY OBJECTS, ONE ANSWER

Choosing the object that matters

$$\text{priority} = 0.50 \times \text{confidence} + 0.35 \times \text{size in frame} + 0.15 \times \text{closeness to centre}$$

Object	Confidence	Size	Centre	Priority
Laptop	0.94	0.42	0.90	0.75 · primary
Bottle	0.86	0.17	0.42	0.55
Chair	0.78	0.24	0.30	0.52

The centre term is intent: you point the phone at what you want to know about.

Left, centre or right comes from where the object sits on screen. We never claim distance without a depth sensor.

Illustrative values, not measurements. All weights and thresholds are configurable and exposed in a developer diagnostics panel.

HOW WE BUILD

Two teams, one app, no collisions

MASTER · MUSIC

Capture, grid, AI Producer, texture

Music files are protected. The vision work never edits them.

FEAT/WORLJDJAM-VISION-READOUT

Detection, tracking, readout, TTS

Separate git worktree, new files only, atomic commits.

THE CONTRACT BETWEEN THEM

VisionEvent { trackId, label, bbox, center, timestamp }

Next phase: object → its captured sound → instrument → AR tile. Vision never imports music code.

PROOF

Measured, not borrowed

What	Result	Where
Stable Audio: one 11-second stereo clip	17.8 s	iQOO 15, 8 threads
Stable Audio peak memory	1.8 GB	iQOO 15
Tap-to-sound latency	24.8 ms	in-app measurement
Gemma 4 E2B model on the phone	3.35 GB	iQOO 15, app storage
Automated tests	481 passing	23 suites
Gemma plan time	to be measured	iQOO 15
YOLO26n inference, fps, thermals	to be measured	iQOO 15, 30 s to 10 min

Rows marked “to be measured” get real iQOO 15 numbers before we quote them. We do not reuse benchmarks from other phones.

ROADMAP

From your room to a shared stage

NOW · BUILT

Music studio

Capture, beat grid,
AI Producer, texture
layer

PHASE 1 · BUILDING

See and say

Object detection
with spoken
readout

PHASE 2

Seen becomes heard

Detected objects
map to their
captured sounds

PHASE 3

AR Tiles

A rhythm game
played on your real
objects

LATER

Together

Multiplayer jams
and a voice agent

Your room is the band.

Real sounds. A brain on the phone. Built for ears first.

Team PRXFR · WorldJam · iQOO City Battles 2026

